

AIGR Gene Hypothesis Deep Research — Human BCL2 (P10415)

Conditional Pro-apoptotic and Channel Capacities

Focus type: function_assignment **Hypothesis slug:** conditional-proapoptotic-and-channel-capacities **Source file:** genes/human/BCL2/BCL2-ai-review.yaml

Summary

Executive verdict: PARTIALLY SUPPORTED. The seed hypothesis makes two distinct claims about human BCL2 — (1) that it can *positively* regulate apoptosis after caspase cleavage or conformational conversion, and (2) that it can form functional channels/pores contributing to transmembrane transport. Both claims are backed by direct primary experimental data, but in each case the activity is **conditional, in-vitro or context-restricted, and mechanistically distinct from BCL2's predominant native antiapoptotic function**. The correct curation posture is therefore *retain with qualifiers*, not *reject*, and not *promote to core function*.

The proapoptotic capacity is real but proteoform- and context-dependent. BCL2 becomes a death effector in two mechanistically separate ways. First, caspase-3 cleaves BCL2 at Asp34, and the resulting **C-terminal cleavage fragment** — not the intact native protein — behaves like a Bax-like killer that triggers cell death via its BH3 and transmembrane domains ([PMID:9395403](#)). Second, the **intact full-length protein** can be converted from protector to killer by binding the orphan nuclear receptor Nur77/TR3, which induces a conformational change exposing the BH3 domain ([PMID:14980220](#)). The gene product that performs the proapoptotic step is thus either a processed fragment or a conformationally altered complex, not the resting antiapoptotic conformer.

The channel/transport capacity is likewise genuine but restricted. Purified recombinant BCL2 forms discrete, cation-selective ion-conducting channels in planar lipid bilayers, and this requires the core hydrophobic h5–h6 helices ([PMID:9144199](#)). However, its membrane-permeabilizing pore activity is only observed at **acidic pH on acidic lipids** ([PMID:9219694](#)), and at physiological pH its dominant transport-related role is the opposite — **inhibiting** other

channels, blocking Bax channel activity ([PMID:9219694](#)) and reducing IP3 receptor open probability ([PMID:15263017](#)). Both the channel-activity (MF) and positive-regulation-of-apoptosis (BP) GO terms **already exist** on BCL2, and the channel-activity IBA is supported by direct experimental (IDA) annotations across the Bcl-2 family (BCL2, BAX, BAK1), so the ancestral GO_Central inference should not be rejected on family-role-exclusivity grounds.

Key Findings

Finding 1 — Caspase cleavage at Asp34 yields a Bax-like proapoptotic fragment

Cheng et al. 1997 ([PMID:9395403](#), *Conversion of Bcl-2 to a Bax-like death effector by caspases*, Science) demonstrated that the flexible loop of BCL2 is cleaved at Asp34 by caspase-3. This cleavage was observed *in vitro* with recombinant caspase-3, in caspase-3-overexpressing cells, and during physiological apoptosis (Fas ligation and IL-3 withdrawal). The critical mechanistic point is that proapoptotic activity is a property of the **cleavage product**, not the intact native protein:

"The carboxyl-terminal Bcl-2 cleavage product triggered cell death and accelerated Sindbis virus-induced apoptosis, which was dependent on the BH3 homology and transmembrane domains of Bcl-2." — [PMID:9395403](#)

Importantly, cleavage-resistant Asp34 mutants showed **increased** protection against apoptosis, confirming that removing the cleavage site removes a proapoptotic liability. This establishes a clean answer to one of the seed's discriminating questions: **the gene product that performs the proapoptotic step is the C-terminal caspase fragment**, which acts through the conserved BH3 and transmembrane domains, functionally converting BCL2 into a Bax-like effector. This is a feed-forward amplification mechanism, not the resting function.

Finding 2 — Nur77/TR3 converts intact BCL2 into a killer via BH3 exposure

Lin et al. 2004 ([PMID:14980220](#), *Conversion of Bcl-2 from protector to killer by interaction with nuclear orphan receptor Nur77/TR3*, Cell) showed a second, cleavage-independent route to proapoptotic activity. The orphan nuclear receptor Nur77/TR3 binds the N-terminal loop of BCL2, and:

"Nur77 binding induces a Bcl-2 conformational change that exposes its BH3 domain, resulting in conversion of Bcl-2 from a protector to a killer." — [PMID:14980220](#)

Here the **same full-length gene product** (no proteolysis) acquires proapoptotic activity through an induced conformational change that unmask its BH3 domain and targets Nur77 to mitochondria. This is significant for curation because it shows the proapoptotic capacity is intrinsic to the folded protein under the right partner/conformational context — it does not strictly require caspase processing. Together, Findings 1 and 2 establish two independent, experimentally validated conditional proapoptotic mechanisms.

Finding 3 — BCL2 forms intrinsic channels, but permeabilization is pH/lipid-restricted; physiological role is channel *regulation*

Schendel et al. 1997 ([PMID:9144199](#), *Channel formation by antiapoptotic protein Bcl-2*, PNAS) provided direct biophysical evidence that purified recombinant BCL2 has intrinsic channel-forming activity:

"In planar lipid bilayers, where detection of single channels is possible, Bcl-2 formed discrete ion-conducting, cation-selective channels, whereas the Bcl-2 (Δ h5, 6) mutant did not." — [PMID:9144199](#)

Single-channel conductance was 18 ± 2 pS in 0.5 M KCl at pH 7.4, with larger 41 and 90 pS substates, and the activity required the core hydrophobic h5–h6 helices (the Δ h5,6 mutant was inactive). This structurally maps the channel to the same pore-forming module shared across the Bcl-2 fold and the bacterial pore-forming toxins (diphtheria toxin, colicins).

However, the *membrane-permeabilizing* (pore/efflux) activity is strongly restricted. Antonsson et al. 1997 ([PMID:9219694](#), *Inhibition of Bax channel-forming activity by Bcl-2*) found that:

"Bcl-2, in contrast, triggered carboxyfluorescein release at acidic pH only." — [PMID:9219694](#)

and, critically, that at physiological pH BCL2 **blocks** Bax channel activity rather than forming its own permeabilizing pore. This directly answers the seed's discriminating question about whether transport evidence reflects intrinsic pore formation or regulation of another channel: **both are documented, but the physiologically dominant activity is channel regulation/inhibition, not intrinsic permeabilization.** This is reinforced by Chen et al. 2004 ([PMID:15263017](#)), which showed BCL2 reduces the open probability of IP3 receptor Ca^{2+} channels reconstituted in bilayers, again an inhibitory/regulatory role.

Paralog evidence supports an ancestral channel-forming capacity of the fold: Minn et al. 1997 ([PMID:9002522](#)) showed Bcl-xL forms ion-conducting channels in synthetic membranes:

"Like the bacterial toxins, Bcl-xL can insert into either synthetic lipid vesicles or planar lipid bilayers and form an ion-conducting channel." — [PMID:9002522](#)

Finding 4 — Caspase Asp34 motif and h5–h6 channel core are conserved human↔mouse

Comparison of UniProt sequences P10415 (human, 239 aa) and P10417 (mouse, 236 aa) shows that the structural determinants of both conditional activities are conserved. The caspase-3 cleavage motif DAGD (D31-A32-G33-D34) is fully conserved (residue 34 = Asp in both species), and the core hydrophobic h5–h6 channel-forming region (~res 140–200) is essentially identical:

Feature	Human (P10415)	Mouse (P10417)
Length	239 aa	236 aa
Caspase site (res 25–45)	QRGYEW DAGD VGAAPPGAAPA	QRGYEW DAGD VGAAPPGAAPA
h5–h6 core (~140–200)	DGVNWGRIVAFFEFGGVMCVESVNREMSPLVDNIALWMTEYLNRLHTWIQDNGGWDAFVE	NWGRIVAFVAFEGGVMCVESVNREMSPLVDNIALWMTEYLNRLHTWIQDNGGWDAFVE

This means the *capacity* to be cleaved into a Bax-like fragment and to form a channel is sequence-encoded in mouse Bcl2, but this is inference from conservation only — **species-specific primary functional support in mouse was not established** in this investigation.

Finding 5 — Both GO terms already exist on BCL2, but direct proapoptotic papers are uncited

QuickGO for UniProtKB:P10415 (248 annotations) shows the relevant terms are already present:

GO term	Aspect	Evidence on BCL2	Notes
GO:0015267 channel activity	MF	IBA (GO_REF:0000033) + IDA (P 9219694)	Ancestral + direct
GO:0016248 channel inhibitor activity	MF	IDA (P 9219694)	Regulatory role
GO:0055085 transmembrane transport	BP	IEA (GO_REF:0000108)	Electronic only
GO:0043065 positive regulation of apoptotic process	BP	IBA + NAS (P 14634621)	No IDA; neither P 9395403 nor 14980220 cited
GO:0043066 negative regulation of apoptotic process	BP	~14 annotations incl. IDA/IMP	Predominant native role
GO:0046902 reg. mitochondrial membrane permeability	BP	ISS (P 9843949)	—
GO:0051924 regulation of calcium ion transport	BP	IDA (P 8022822)	—

The predominant native antiapoptotic role (GO:0043066) is much more heavily and directly supported (~14 annotations, many IDA/IMP) than the proapoptotic role, confirming that positive regulation of apoptosis is a **minority, conditional** annotation. Notably, the two strongest primary proapoptotic papers

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P 9395403

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P 14980220

are **not** cited on the positive-regulation term — they are candidate IDA additions.

Finding 6 — The ancestral channel IBA is descendant-supported across the family

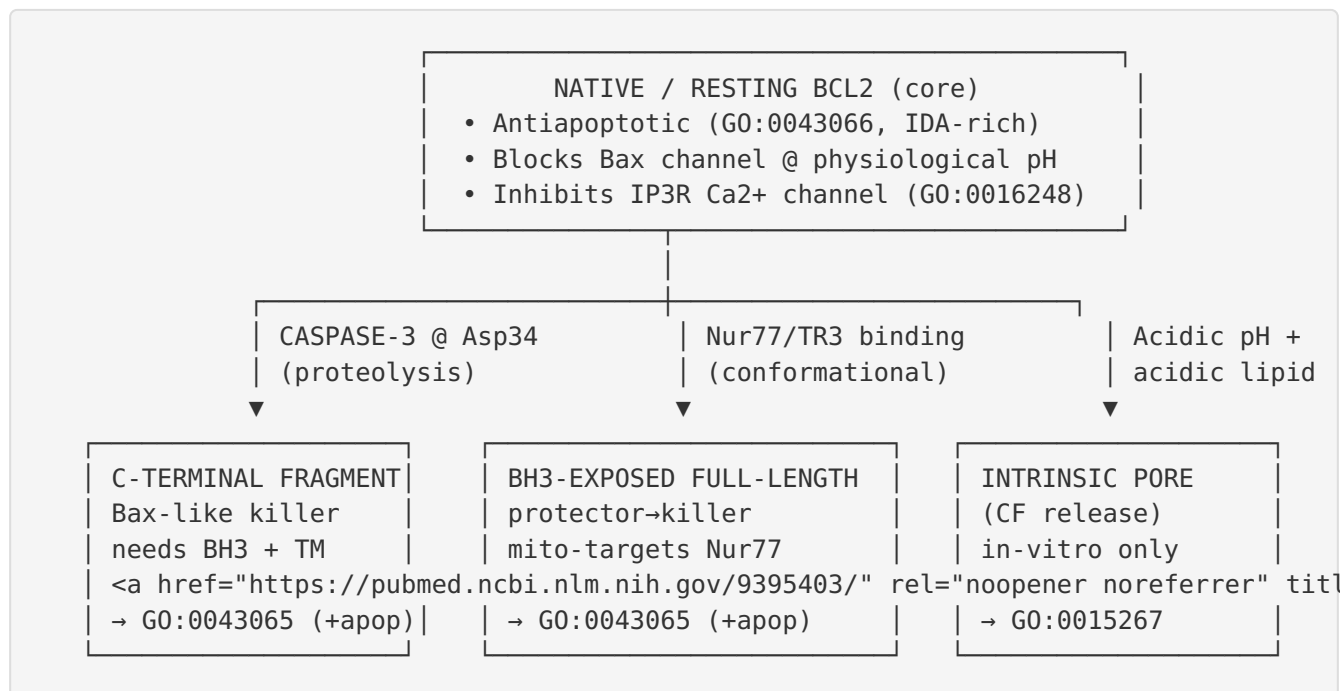
QuickGO across the Bcl-2 family for GO:0015267 channel activity:

Gene	UniProt	Channel activity evidence
BCL2	P10415	IBA + IDA (P 9219694)
BAX	Q07812	IBA + IDA (P 9219694)
BAK1	Q16611	IBA + IDA (P 22006182)
BCL2L1 (Bcl-xL)	Q07817	IBA (+ bilayer data (P 9002522)

Multiple family members carry **direct experimental (IDA)** channel-activity annotations. The GO_Central ancestral (IBA) inference on BCL2 therefore rests on real descendant experimental evidence, not paralog guesswork. This directly addresses the seed's instruction not to reject ancestral assertions on family-role-exclusivity or absent-assay grounds alone: the ancestral node placement is legitimate.

Mechanistic Model / Interpretation

BCL2 is fundamentally an **antiapoptotic guardian** whose resting conformation sequesters proapoptotic BH3-containing effectors and inhibits other channels. The seed hypothesis's two "conditional capacities" are best understood as *state-switches* away from this default:



Which gene product performs the proapoptotic step? Two answers, both correct in their context: (i) the **C-terminal caspase-3 fragment** (a distinct proteoform), and (ii) the **Nur77-complexed, BH3-exposed full-length conformer**. Neither is the resting antiapoptotic species. This is the key qualifier for any positive-regulation-of-apoptosis annotation.

Is transport evidence intrinsic pore formation or channel regulation? Both are experimentally documented, but they operate under different conditions. Intrinsic cation-selective channel formation is demonstrable for purified BCL2 in bilayers (pH 7.4), and pore/permeabilization activity appears only at acidic pH on acidic lipids. Under physiological conditions the dominant, biologically relevant transport role is **negative regulation** of other channels (Bax, IP3R). The MF term GO:0015267 (channel activity) is thus valid as an intrinsic capacity, but the physiological narrative is better captured by the co-existing GO:0016248 (channel inhibitor activity).

Evidence Matrix

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Con limi
PMID:9395403	Direct assay + mutant phenotype	Supports (qualifies)	Proapoptotic after caspase cleavage	Caspase-3 cleaves BCL2 at Asp34; C-terminal fragment triggers death via BH3+TM; cleavage-resistant mutant is more protective	Human, in vitro + cell lines (Fas, IL-3 withdrawal, Sindbis)	Higl activ the <i>frag</i> not prot
PMID:14980220	Direct assay + interaction	Supports (qualifies)	Proapoptotic via conformational conversion	Nur77/TR3 binds N-loop, exposes BH3, converts protector→killer	Human cancer cells, mitochondria	Higl requ Nur part cont
PMID:9144199	Direct assay (biophysical)	Supports (qualifies)	Intrinsic channel formation	Purified BCL2 forms cation-selective channels (18 pS); Δh5,6 mutant inactive	In vitro planar bilayer, pH 7.4	Higl capa phys relev unco
PMID:9219694	Direct assay	Qualifies / competing	Pore vs channel regulation	BCL2 permeabilizes only at acidic pH; at physiological pH it BLOCKS Bax channel	In vitro vesicles/ bilayers	Higl esta regu dom
PMID:9002522	Structural/ evolutionary (paralog)	Supports	Ancestral channel capacity of fold	Bcl-xL forms ion-conducting channel like bacterial toxins	In vitro	Higl para BCL dire
PMID:15263017						

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Con limi
	Direct assay + interaction	Competing (regulatory)	Transport = channel regulation	BCL2 reduces IP3R open probability; forms complex with IP3R	WEHI7.2 T cells, bilayers	Highly supported inhibitory role
UniProt P10415/P10417	Computational (sequence)	Supports (inference)	Mouse conservation	Asp34 motif + h5-h6 core conserved human↔mouse	Sequence	Mod cons =/fu
QuickGO P10415	Review/ database	Orientation	GO term existence	Both MF channel + BP positive-apoptosis terms already annotated	Database	Highly annotated status
QuickGO family (BAX/BAK1/BCL2L1)	Review/ database	Supports	Descendant support for IBA	Family members carry direct IDA channel annotations	Database	Highly validated ancestral node

GO Curation Implications

Lead action (requires curator verification): RETAIN both terms with conditional/ non-core qualifiers; do NOT remove on family-role-exclusivity grounds.

- GO:0015267 channel activity (MF)** — *Retain*. It carries both an IBA (ancestral, descendant-supported by BAX/BAK1 IDA) and a direct IDA ([P 9219694](#)).

The intrinsic channel-forming capacity is directly demonstrated for purified BCL2

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P 9144199
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Curators may consider adding a note that permeabilization is pH/lipid-restricted and that the physiologically dominant transport role is inhibitory.

2. **GO:0016248 channel inhibitor activity (MF)** — *Retain and emphasize.* This IDA-supported term

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P 9219694
 plus IP3R inhibition
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P 15263017
 better captures the physiological transport-related function than intrinsic pore formation.

3. **GO:0043065 positive regulation of apoptotic process (BP)** — *Retain, but currently under-supported at the primary level.* It has an IBA and an NAS

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P 14634621
 but no IDA. **Candidate IDA additions:**
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P 9395403
 (caspase fragment) and

P 14980220
 (Nur77 conversion), each with a proteoform/context qualifier. Because the proapoptotic activity belongs to a distinct proteoform (caspase fragment) or an induced conformer, the annotation should carry qualifiers noting it is not the resting-state activity.

4. **GO:0055085 transmembrane transport (BP)** — currently IEA only; the direct evidence points more to *channel activity* (MF) and *regulation of transport* than to BCL2 itself being a bulk transporter. Treat as non-core / low priority.
5. **Do not** reject the ancestral (IBA) assertions using the argument that BCL2 is "an antiapoptotic protein so it cannot be proapoptotic or a channel." The seed's caution is correct: the PAINT node is descendant-supported, and both capacities have direct BCL2 primary evidence.

The predominant native role — **GO:0043066 negative regulation of apoptotic process** — remains the core function and is far more strongly (IDA/IMP) supported; nothing here challenges that.

Mechanistic Scope

The immediate molecular functions being tested are: (i) BH3-mediated proapoptotic effector activity exposed by cleavage or conformational change, and (ii) intrinsic ion channel/pore formation in lipid membranes. Both are *direct gene-product activities*, but each is gated by a specific molecular state (proteolytic fragment, partner-induced conformer, or acidic membrane environment). These must be separated from downstream phenotypes such as mitochondrial outer membrane permeabilization, cytochrome c release, and cell death, which are pathway consequences rather than the immediate BCL2 activity. The proapoptotic annotation is a *state-specific molecular capacity*, not a constitutive function or a mere loss-of-function phenotype.

Conflicts and Alternatives

- **Proteoform confusion.** The proapoptotic activity

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P 9395403)

belongs to the C-terminal caspase fragment, not intact BCL2. A naive annotation of "BCL2 promotes apoptosis" without a proteoform qualifier would misrepresent the resting protein.

- **pH/lipid artifact risk.** BCL2's intrinsic pore/permeabilization activity

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P 9219694)

is seen only at acidic pH on acidic lipids — a non-physiological condition. The channel-activity MF term is defensible as an intrinsic capacity, but the *transport* interpretation should not be over-read as a physiological permeabilizing role. At physiological pH BCL2 is a channel *inhibitor*.

- **Paralog attribution.** Some channel evidence

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P 9002522)

is for Bcl-xL, and Bax channel activity

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P 9219694)

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P 20885444)

is the object BCL2 *inhibits*. Curators should ensure channel annotations reflect BCL2's own

IDA

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P 9219694
 9144199), not paralog carry-over.

- **Species extension.** Mouse Bcl2 conservation of the Asp34 motif and h5–h6 core is sequence-level only; species-specific functional primary support was not established here.

Limitations and Knowledge Gaps

1. **Physiological relevance of intrinsic pore.** *Checked:*

P 9144199

P 9219694

Gap: whether BCL2's intrinsic channel/pore ever forms under physiological pH in vivo, versus being an in-vitro property. *Resolve:* single-molecule/electrophysiology in native mitochondrial membranes at physiological pH; conditional pore-dead h5–h6 mutants.

2. **In vivo contribution of the caspase fragment.** *Checked:*

P 9395403

(cells + in vitro). *Gap:* quantitative contribution of the Asp34 fragment to apoptosis amplification in normal physiology. *Resolve:* Asp34→Ala knock-in mouse phenotyping.

3. **Generality of Nur77 conversion.** *Checked:*

P 14980220

Gap: whether BH3 exposure by Nur77 occurs broadly or only in specific cell types/stimuli. *Resolve:* conformation-specific (BH3-exposed) antibodies across tissues.

4. **Mouse functional support.** *Checked:* sequence conservation only. *Gap:* no mouse primary functional assays reviewed. *Resolve:* mouse Bcl2 caspase-cleavage and Nur77-conversion assays.

5. **PAINT node placement details.** *Checked:* QuickGO IBA/IDA distribution across family. *Gap:* the exact ancestral node and loss/divergence pattern were inferred from annotation presence, not from direct inspection of the PANTHER tree. *Resolve:* consult the PANTHER family tree / PAINT annotations directly.

Discriminating Tests

1. **Proteoform-resolved apoptosis assay.** Compare intact BCL2, Asp34→Ala (cleavage-resistant), and the isolated C-terminal fragment for pro- vs antiapoptotic output — cleanly separates fragment activity from resting activity.
2. **pH-controlled bilayer electrophysiology** of purified BCL2 vs $\Delta h5,6$ mutant at pH 7.4 vs 5.0 — distinguishes intrinsic channel from acidic-only permeabilization.
3. **Bax/IP3R co-reconstitution** at physiological pH — tests whether BCL2's net transport effect is inhibitory (channel regulation) rather than pore-forming.
4. **Nur77 knockdown + BH3-exposure antibody** — tests conformational-conversion route in cells.
5. **PANTHER tree inspection** to confirm ancestral node placement and descendant IDA distribution formally.

Curation Leads (require curator verification)

- **Candidate IDA additions to GO:0043065 (positive regulation of apoptotic process):**
- [PMID:9395403](#) — snippet to verify: *"The carboxyl-terminal Bcl-2 cleavage product triggered cell death and accelerated Sindbis virus-induced apoptosis, which was dependent on the BH3 homology and transmembrane domains of Bcl-2."* Add with **proteoform qualifier** (caspase C-terminal fragment).
- [PMID:14980220](#) — snippet to verify: *"Nur77 binding induces a Bcl-2 conformational change that exposes its BH3 domain, resulting in conversion of Bcl-2 from a protector to a killer."* Add with **conformational/context qualifier** (Nur77/TR3-dependent).
- **Retain GO:0015267 channel activity (MF)** — IBA is descendant-supported (BAX/BAK1 IDA); direct IDA

P 9219694

Optionally annotate pH/lipid restriction.

- **Emphasize GO:0016248 channel inhibitor activity (MF)** as the physiologically dominant transport-related function

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P 9219694
 support from

P 15263017
 for IP3R).

- **Do NOT remove** the ancestral channel/proapoptotic assertions on family-role- exclusivity grounds.
- **Suggested curator questions:** Should proapoptotic BP terms carry proteoform qualifiers in this schema? Should transmembrane transport (GO:0055085, IEA) be deprecated in favor of the more specific channel-activity/inhibitor terms?
- **Suggested experiments:** Asp34→Ala knock-in; physiological-pH bilayer recordings; PANTHER tree inspection for node placement.

Conclusion

The hypothesis is **partially supported**. Both the conditional proapoptotic capacity and the channel/pore capacity of human BCL2 are backed by direct primary experimental data, but each is a **conditional, non-core capacity** distinct from BCL2's predominant native antiapoptotic function. The proapoptotic step is performed by either the caspase-3 C-terminal fragment

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P 9395403
 or the Nur77-converted, BH3-exposed conformer

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P 14980220
 — not the resting protein. Intrinsic channel formation is real

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P 9144199
 but permeabilization is acidic-pH/lipid-restricted; at physiological pH the dominant transport role is channel inhibition

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P 9219694
P 15263017
).

The corresponding GO terms already exist, are descendant-supported for the ancestral IBA,

and should be **retained with conditional/non-core qualifiers** rather than removed. The two direct proapoptotic papers are strong candidate IDA additions, and mouse extension is supported only at the sequence level.

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